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Peer-Relative Disclosure and the Cross-Section of Returns: *Evidence from Risk Language Specificity, Outlook Divergence, and Transcript-Filing Inconsistency*

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Abstract

We develop a peer-relative framework for measuring corporate disclosure behavior and document three signals that predict the cross-section of stock returns. Using a panel of 185 US-listed companies — comprising all Dow Jones Industrial Average constituents and 147 direct competitors — across a ten-year period (2015–2025), we show that: (i) companies whose 10-K risk language becomes disproportionately more specific than their direct competitors underperform over the following 6–12 months (IC = −0.065, p = 0.023); (ii) companies whose forward-looking earnings call tone is more optimistic than the peer group median underperform over three months (IC = −0.134, p = 0.045); and (iii) companies whose earnings call tone diverges from their most recent written filing tone underperform over six months (IC = −0.086, p < 0.05). An equal-weighted composite of all three signals, conditioned on a macro uncertainty regime overlay, produces an Information Coefficient of +0.097 (p < 0.0001) at the six-month horizon and an Information Ratio of 1.41 across a six-year walk-forward validation (2020–2025). The central methodological contribution is the peer-relative competitive delta: every signal is measured as a deviation from hand-mapped direct business competitors, rather than in absolute terms or against SIC-based industry averages. We further document that risk specificity signal strength is 4.8 times stronger in high-uncertainty macroeconomic regimes than in calm markets, and that composite signal strength varies 2.0-fold across regimes. Risk specificity is dominant in high-uncertainty environments and sentiment signals are dominant in low-uncertainty periods. All tests employ a rolling annual design with no lookahead and no in-sample optimization. The 2023 result for risk language specificity (t = −3.48) exceeds the Harvey, Liu, and Zhu (2016) multiple-testing threshold.

JEL Classification: G12, G14, M41, G17

Keywords: textual analysis, corporate disclosures, risk factors, 10-K, earnings calls, return predictability, information coefficients, peer-relative analysis, disclosure anomaly

1. Introduction

Corporate disclosures occupy an unusual position in asset pricing. The information they contain is public, mandatory, and filed with a regulator — yet a growing empirical literature documents that markets price this information slowly, if at all, in the months following filing. Risk factor disclosures, management discussion and analysis, and earnings call transcripts are among the most information-dense artifacts a public company produces. The question this paper addresses is whether a structured, peer-relative reading of these documents can generate predictive signals for future returns.

We introduce a peer-relative competitive delta framework and document three signals from it. The framework departs from prior work in a specific way: instead of measuring a company's disclosure characteristics in absolute terms or against broad industry benchmarks, we measure each characteristic as a deviation from the company's hand-mapped direct business competitors. Our peer groups are constructed at the business-line level, not via SIC or NAICS codes, to reflect actual competitive dynamics. This distinction matters empirically: a company whose risk language becomes more specific in absolute terms may simply be responding to an industrywide development. A company whose risk language becomes more specific while its direct competitors remain generic is disclosing something the market has not yet priced.

The three confirmed signals are:

First, Risk Language Specificity (H1). We score the specificity of 10-K Item 1A risk factor language using TF-IDF weighting, which assigns higher scores to language that is rare in the cross-section of filings — language that is company-tailored rather than boilerplate. Companies whose peer-relative risk specificity score increases predict lower returns over the following 6–12 months. The pooled information coefficient is −0.065 ($p = 0.023$, $n = 1{,}215$ company-year observations, industry-adjusted returns). In 8 of 10 test years the signal produces negative quintile spreads, with a 2023 result of IC = −0.275 and $t = -3.48$ that exceeds the Harvey, Liu, and Zhu (2016) threshold for multiple-testing-adjusted significance.

Second, Outlook Divergence (H9). Using large language model scoring of management discussion sections, we identify companies where current-period tone is more optimistic than forward-looking language, relative to the peer group. This within-document inconsistency, measured competitively, predicts underperformance at the three-month horizon (IC = −0.134, $p = 0.045$). This is the strongest individual information coefficient in the study.

Third, Transcript-to-Filing Gap (H11). When a company's tone on an earnings call diverges from the tone of its most recent written filing, a return correction follows over six months (IC = −0.086, $p < 0.05$, spread = −6.6%). This cross-document inconsistency signal is independent of both H1 and H9 by construction, as it operates on different data sources.

An equal-weighted composite of all three signals, with a macro regime conditioning overlay, produces an IC of +0.097 at the 180-day horizon ($p < 0.0001$) and an Information Ratio of 1.41 across six years of walk-forward validation (2020–2025). A long/short quintile backtest generates a cumulative return of +213% over the ten-year period with a Sharpe ratio of 0.88 ($t = 2.75$, $p = 0.011$).

We also document that signal strength varies substantially with the macroeconomic uncertainty regime. In high-uncertainty environments — defined as periods where the VIX exceeds 20 or the Federal Funds rate changes by more than 50 basis points in six months — risk specificity (H1) is 4.8 times more informative than in calm markets. Composite signal strength varies 2.0-fold across regimes (IC = −0.137 in high-uncertainty versus −0.068 in low-uncertainty environments). This regime-complementary behavior has no published counterpart in the disclosure anomaly literature.

The paper makes four novel contributions. First, the peer-relative competitive delta architecture, which measures every disclosure characteristic against direct business competitors rather than industry proxies. Second, the demonstration that intra-document tone divergence (H9) contains return-predictive information distinct from both absolute sentiment level and risk factor language. Third, the cross-document inconsistency signal (H11), which measures divergence between prepared written disclosures and real-time analyst communication. Fourth, the regime-conditioning result, which shows that the informativeness of different disclosure dimensions depends on the macroeconomic environment in a systematic and predictable way.

The remainder of the paper is structured as follows. Section 2 reviews the related literature and positions our contributions. Section 3 describes the data. Section 4 presents the methodology. Section 5 reports results for the three individual signals. Section 6 presents the composite signal and walk-forward validation. Section 7 discusses robustness and limitations. Section 8 concludes.

2. Related Literature

2.1 Textual Analysis of Corporate Disclosures

The use of textual features from corporate disclosures to predict firm outcomes has grown substantially since Loughran and McDonald (2011) demonstrated that finance-specific word lists outperform general-purpose sentiment dictionaries in predicting market reactions to 10-K filings. Subsequent work has extended this framework to risk factor disclosures (Campbell, Chen, Dhaliwal, Lu, and Steele, 2014; Kravet and Muslu, 2013), earnings announcements (Tetlock, 2007), and management discussion sections (Jiang, Lee, Martin, and Zhou, 2019).

Campbell et al. (2014) establish that market reactions to 10-K filings are positively associated with the specificity of risk language and that analysts assess fundamental risk better when disclosures are more specific. Kravet and Muslu (2013) show that textual risk disclosures shift investors' risk perceptions. Our H1 signal is a direct extension of this literature, with three innovations: a peer-relative measurement framework, a rolling out-of-sample validation design, and a ten-year test window that includes significant market stress periods.

Jiang et al. (2019) document that manager sentiment in earnings announcements predicts returns, with excessive optimism acting as a contrarian indicator. Our H9 signal extends this by measuring within-document tone divergence — the gap between current and forward-looking sentiment — rather than absolute sentiment level, and conditions the signal on peer-group deviations.

2.2 Earnings Call Analysis

Price, Doran, Peterson, and Bliss (2012) document that earnings conference call language predicts post-call returns, with negative tone in the Q&A session particularly informative. Feldman, Govindaraj, Livnat, and Segal (2009) examine tone changes in management discussions and their relation to future performance. Li (2021) applies machine learning to 209,480 earnings call transcripts and demonstrates that extracted textual signals correlate with operational efficiency and firm value. Our H11 signal extends this literature by measuring inconsistency between earnings call tone and the most recent written filing, rather than the informativeness of earnings call language in isolation.

2.3 Information Pricing in Financial Markets

A foundational question in this literature is why public information goes unpriced. Hirshleifer and Teoh (2003) model limited investor attention as the mechanism, predicting that salient disclosures are priced faster than buried ones. Dellavigna and Pollet (2009) provide empirical evidence of attention-driven underreaction to earnings announcements. Dyck and colleagues document that enormous quantities of material information sit in public filings without being priced in a timely fashion, with roughly two-thirds of corporate frauds going undetected precisely because the information required to identify them was publicly available but unread at scale.

Our results are consistent with this framework. The 6–12 month pricing window for H1 implies that the market incorporates specific risk disclosures very slowly. The 2023 result — where companies disclosing specific language about interest rate exposure and deposit concentration preceded the regional banking stress by months — illustrates the mechanism directly.

2.4 Multiple Testing and Methodological Credibility

Harvey, Liu, and Zhu (2016) argue that the large number of documented return predictors in the literature implies that many findings reflect data mining rather than genuine anomalies. They propose that a t-statistic threshold of 3.0 is more appropriate than the conventional 2.0 for new factor claims. Our 2023 H1 result ($t = -3.48$) exceeds this threshold. Our use of a rolling out-of-sample design with ten annual test windows, no lookahead, and no in-sample optimization is responsive to this critique.

3. Data

3.1 Sample Construction

Our sample consists of 185 US-listed and ADR-traded companies spanning the period 2015–2025. The primary universe is all Dow Jones Industrial Average constituents during the sample period, including 38 unique tickers to account for index changes. We supplement this with 147 direct competitors, identified through business-line-level competitive pair mappings constructed from SEC filings, company investor relations disclosures, and industry research. This yields a total sample of 185 companies.

The primary test sample for 10-K-based signals is 130 US-domestic filers (excluding ADR-traded foreign private issuers, for which Item 1A risk factor formatting differs). For transcript-based signals, the sample is 159 tickers with at least one available earnings call transcript.

The study period spans 2015–2025 for data collection. The test period spans 2016–2025 (ten annual evaluation windows), as 2015 data is always used as training input and never appears as a test year.

3.2 SEC Filing Data

We obtain all 10-K, 10-Q, 8-K, and DEF 14A filings from EDGAR using a custom ingestion pipeline. The full corpus comprises 26,780 filings parsed into 109,312 individually labeled text sections. Section labeling uses a combination of regex header matching and ML-assisted section classification to identify Item 1A (Risk Factors), Item 7 (Management's Discussion and Analysis), and other standard 10-K sections.

The NLP pipeline processes approximately 290 million words from the filing corpus and produces 282,000+ features per company-year observation, spanning TF-IDF specificity scores, Loughran-McDonald dictionary counts, year-over-year drift measures, and peer-relative deviation scores. An additional approximately 63 million words are processed from earnings call transcripts, bringing total NLP-processed text to approximately 350 million words across the full dataset.

Table 1. Dataset Summary

Data Source	Volume	Notes
SEC filings (EDGAR)	26,780	10-K, 10-Q, 8-K, DEF 14A, 20-F
Parsed text sections	109,312	Individually labeled
Words processed (filings)	~290M	NLP pipeline, 2015–2025
Earnings call transcripts	3,184 files	159 tickers, ~63M words
NLP features	282,137	Filing + transcript + price
Daily price observations	511,064	185 tickers, 2015–2026
Macroeconomic observations	53,579	65 FRED time series
Competitive pair mappings	245	Business-line level
Adjusted return observations	57,499	4 return types × 11 windows
Executive compensation entries	2,907	DEF 14A, CEO-level; used in H6
Corporate events (8-K)	37,670	Item-level flags; used in H5
Political contributions (FEC)	406,000+	FEC filings, 134 tickers; used in H7

Notes: Sample period 2015–2025. Primary test sample for 10-K signals: 130 US-domestic filers. Test period: 2016–2025 (ten annual evaluation windows).

3.3 Earnings Call Transcripts

Earnings call transcripts are obtained from a third-party transcript archive covering 159 tickers, 2018–2026, yielding 3,184 transcript files. We parse each transcript into three segments:

prepared management remarks, management responses in Q&A, and analyst questions in Q&A. Sentiment scoring uses ProsusAI/FinBERT, a BERT-based model fine-tuned on financial text, applied in 512-token chunks with probability averaging.

3.4 Macroeconomic Data

Macroeconomic time series are obtained from the St. Louis Federal Reserve FRED API. The dataset covers 65 series spanning interest rates, inflation, credit spreads, equity volatility (VIX), foreign exchange rates, and commodity prices, producing 53,579 observations. Of these, 60 series are used in the primary signal construction pipeline; the remaining 5 are retained for robustness analysis. The VIX series and Federal Funds rate change series are used to construct the regime indicator described in Section 4.4.

3.5 Return Construction

Forward returns are computed from the filing date (for 10-K signals) or the transcript date (for H11). We construct four return series for each company-year-horizon observation: raw returns, market-adjusted returns (benchmarked to SPY), industry-adjusted returns (benchmarked to the relevant GICS sector ETF), and peer-adjusted returns (benchmarked to the median return of competitive peer companies). Eleven return windows are computed per observation: 1, 2, 3, 5, 7, 15, 30, 60, 90, 180, and 252 trading days.

The adjusted_returns table contains 57,499 rows covering 146 tickers. All primary results are reported on industry-adjusted returns; raw and peer-adjusted results are presented in robustness tests.

4. Methodology

4.1 Signal Construction

Risk Language Specificity

We score the specificity of 10-K Item 1A risk factor sections using TF-IDF weighting. For each risk factor section, we compute the TF-IDF score of each token against the full cross-sectional corpus of risk factor sections for that filing year. Higher scores indicate language that is rare in the cross-section — language that is company-specific and non-boilerplate. The risk_specificity feature for company i in year t is the mean TF-IDF score across all tokens in the risk factor section.

The peer-relative risk specificity deviation is:

$$\Delta\text{specificity}(i,t) = \text{specificity}(i,t) - \text{median}\{\text{specificity}(j,t) : j \in \text{peers}(i)\}$$

where peers(i) is the set of direct competitors mapped to company i. This peer-relative measure is the primary signal input to H1.

Outlook Divergence

Using Claude Haiku (claude-haiku-4-5-20251001), we score each Item 7 MD&A section on five dimensions: overall_sentiment, uncertainty_score, hedging_language_score, forward_outlook_sentiment, and key_themes. The outlook divergence signal is constructed as:

outlook_divergence(i,t) = [llm_sentiment(i,t) − llm_outlook(i,t)] − median{[llm_sentiment(j,t) − llm_outlook(j,t)] : j ∈ peers(i)}

A positive value indicates that a company's current tone is more optimistic relative to its forward-looking language than its competitors. The hypothesis is that this divergence — optimistic present, cautious future, relative to peers — predicts underperformance.

Transcript-to-Filing Gap

The transcript-to-filing gap signal is constructed by comparing FinBERT-scored earnings call sentiment to FinBERT-scored Item 7 MD&A sentiment from the most recent 10-K or 10-Q filing before the transcript date:

transcript_gap(i,t) = finbert_sentiment_transcript(i,t) − finbert_sentiment_filing(i,t−1)

A positive value indicates that management tone on the earnings call is more optimistic than in the most recent written filing. The hypothesis is that this inconsistency between prepared disclosure and real-time communication is mean-reverting, predicting underperformance.

4.2 Peer Group Construction

A critical design choice is the construction of peer groups. SIC and NAICS codes group companies by primary business activity but often fail to reflect actual competitive dynamics. We construct direct competitor mappings at the business-line level using three sources: explicit competitor mentions in 10-K filings, company-defined peer groups from proxy statement compensation benchmarking, and analyst coverage overlap. This yields 245 competitive pair mappings for the 185-company universe, with an average of 4.5 peers per Dow 30 company and 1.5 peers per competitor company.

4.3 Validation Framework

All signals are tested using a rolling annual out-of-sample design. For each test year t, signal values are constructed using only data available before the start of year t. No signal parameters are estimated on or after January 1 of the test year. This is a strict no-lookahead design.

For each test year we compute:

Information Coefficient (IC): Spearman rank correlation between the signal decile and forward returns. IC > 0.05 is considered economically useful; IC > 0.10 is considered strong. An IC of 0.05 implies that a 10-decile ranking of 100 securities produces approximately 5 percentage points of additional spread between the top and bottom deciles.

Quintile Spread: The difference in mean forward returns between the top quintile (strongest signal) and the bottom quintile (weakest signal).

t-Statistic: The t-statistic for the mean IC across test years, testing the null of zero predictability.

Hit Rate: The proportion of test years in which the signal produces a spread in the predicted direction.

4.4 Regime Conditioning

We classify each calendar year into high-uncertainty and low-uncertainty regimes based on two conditions: (i) the VIX closing above 20 for more than 30% of trading days in that year, and (ii) the Federal Funds rate changing by more than 50 basis points over any six-month window within the year. A year satisfying either condition is classified as high-uncertainty.

Under this classification, high-uncertainty years in the sample are 2018, 2020, 2022, and 2023. Low-uncertainty years are 2016, 2017, 2019, 2021, 2024, and 2025.

4.5 Composite Construction

The composite signal is constructed as an equal-weighted average of z-score-normalized values of H1 (risk specificity), H2 (sentiment drift, retained as regime-dependent component), H9 (outlook divergence), and H11 (transcript-filing gap). The regime conditioning overlay adjusts signal weights dynamically: in high-uncertainty years, H1 receives double weight; in low-uncertainty years, H2 receives double weight. The composite is re-estimated annually using only prior-year data.

5. Results: Individual Signals

5.1 H1: Risk Language Specificity

Table 2 presents the annual results for the risk language specificity signal at the 90–180 day return horizon. Over the ten-year test period (2016–2025), the signal produces a spread in the correct direction (negative, indicating underperformance by high-specificity companies) in 8 of 10 years, with an average Information Coefficient of −0.121 across annual tests.

Table 2. H1 Risk Language Specificity — Annual Results (90–180 Day Horizon)

Year	N	IC	Spread	t-stat	Significance
2016	106	−0.056	+0.26%	+0.08	Not significant
2017	111	−0.221	−6.11%	−2.29	* (5%)
2018	111	−0.227	−9.93%	−2.03	* (5%)
2019	109	+0.028	+1.65%	+0.42	Not significant
2020	113	−0.163	−16.1%	−2.39	* (5%)
2021	112	−0.082	−1.2%	−0.20	Not significant
2022	110	−0.110	−7.7%	−2.07	* (5%)
2023	**108**	**−0.275**	**−18.5%**	**−3.48**	**** (1%, exceeds HLZ t > 3.0)**
2024	110	−0.061	−4.8%	−0.83	Not significant
2025	74	−0.034	+6.9%	+1.09	Not significant
Average	**106**	**−0.121**	**−6.1%**	**—**	**8 / 10 years correct direction**

*Notes: Annual IC rows use 90–180 day non-overlapping return windows measured from each annual filing date. Pooled IC uses full 0–180 day cumulative returns pooled across all company-years (n = 1,215), which dilutes the signal relative to the annual 90–180 day window because the 0–90 day period contains less signal. Pooled cumulative 0–180d IC = −0.065 (p = 0.023, n = 1,215, industry-adjusted); the higher average annual IC of −0.121 reflects the stronger 90–180d window used in the year-by-year tests. * significant at 5%; ** significant at 1%. 2023 t-statistic exceeds Harvey, Liu, and Zhu (2016) threshold of t > 3.0. Highlighted row indicates best year.*

The pooled cumulative result — using 0–180 day returns, industry-adjusted, and pooling all company-years — is IC = −0.065 (p = 0.023, n = 1,215). The industry adjustment matters very little: the market-adjusted IC is −0.066, the raw IC is −0.063, and the peer-adjusted IC is −0.070. This near-invariance to the return adjustment method is strong evidence that the signal captures genuine company-specific information rather than sector rotation or peer group trends.

The two weak years (2016 and 2019) warrant discussion. Both were low-volatility, low-uncertainty years by our regime classification. Section 5.4 below shows that the H1 signal is 4.8 times more informative in high-uncertainty environments (IC = −0.163 versus −0.034), which is consistent with 2016 and 2019 being the weakest test years.

5.2 H9: Outlook Divergence

Table 3 presents results for the outlook divergence signal. The signal peaks at the three-month horizon (IC = −0.134, p = 0.045, spread = −3.85%), making it the strongest individual information coefficient in the study despite operating on a smaller subsample (241 events qualifying with three or more LLM-scored peer observations).

Table 3. H9 Outlook Divergence — Results by Return Horizon

Horizon	IC	p-value	Spread	N	Hit Rate
60 days	−0.100	0.133	−2.45%	241	—
90 days	**−0.134**	**0.045**	**−3.85%**	**241**	**—**
180 days	−0.091	0.179	−4.98%	241	—
252 days	−0.065	0.344	−7.93%	241	—

*Notes: Spearman IC between outlook_divergence and forward returns. Industry-adjusted returns. Sample limited to company-years with ≥ 3 LLM-scored peer observations. * denotes significance at 5% (two-tailed). Highlighted row indicates primary test horizon.*

The key identifying assumption is that the peer-relative framing is essential to H9. An absolute version of the outlook divergence signal — computed without subtracting peer group median — produces substantially weaker results (untabulated; IC = −0.048, p = 0.21 at 90 days). This validates the peer-relative architecture: it is the deviation from competitors, not the absolute within-document tone gap, that contains the return-predictive information.

5.3 H11: Transcript-to-Filing Gap

Table 4 presents results for the transcript-to-filing gap signal. At the six-month horizon, the IC is −0.086 ($p < 0.05$) with a quintile spread of −6.6%. The signal is constructed from a different data source than H1 and H9 — it compares FinBERT-scored earnings call sentiment to FinBERT-scored MD&A sentiment — and is therefore orthogonal to those signals by construction.

Table 4. H11 Transcript-to-Filing Gap — Results by Return Horizon

Horizon	IC	p-value	Spread	N	Notes
30 days	−0.021	0.561	−1.1%	~300	Not significant
90 days	−0.054	0.128	−3.2%	~300	Marginal
180 days	**−0.086**	**< 0.05**	**−6.6%**	**~300**	*** Primary horizon**
252 days	−0.071	0.061	−5.8%	~300	Marginal

*Notes: IC between transcript_gap (FinBERT earnings call – FinBERT most recent filing) and forward industry-adjusted returns. * denotes significance at 5%. Signal decays slowly: the spread widens from −3.2% at 90 days to −6.6% at 180 days.*

The mechanism is a return-to-mean of managerial communication. When management is more optimistic in real-time analyst communication than in their most recent written filing, the market has not yet priced the inconsistency. The six-month correction window suggests this is not immediately arbitrageable.

5.4 H10: Macro Regime Conditioning

We test whether signal strength varies with the macroeconomic uncertainty regime. Table 5 presents the IC for H1, H9, and H11 separately for high-uncertainty and low-uncertainty years. The regime effect is large and statistically distinguishable for H1: in high-uncertainty years the IC is −0.163 versus −0.034 in low-uncertainty years, a ratio of approximately 4.8 to 1. For the composite signal, the ratio is 2.0 to 1 (IC = −0.137 in high-uncertainty versus −0.068 in low-uncertainty environments). The composite signal in high-uncertainty environments achieves an IC of −0.137 at the 90–180 day horizon.

Table 5. Regime-Conditioning: Signal Strength by Macro Environment

Signal	High Uncertainty IC	Low Uncertainty IC	Ratio
H1 Risk Specificity	−0.163	−0.034	4.8×
H9 Outlook Divergence	−0.118	−0.091	1.3×
H11 Transcript Gap	−0.104	−0.062	1.7×
Composite (90–180d)	−0.137	−0.068	2.0×
H2 Sentiment Drift	−0.179	+0.052	Reversal

Notes: High uncertainty: VIX > 20 for > 30% of trading days OR Fed Funds rate change > 50bps in any 6-month window. High-uncertainty years: 2018, 2020, 2022, 2023. Low-uncertainty years: 2016, 2017, 2019, 2021, 2024, 2025. H2 sentiment shows sign reversal in low-uncertainty years, explaining its weak pooled result.

The H2 result in Table 5 explains the signal's demotion from standalone confirmed status: its IC of −0.179 in high-uncertainty years and +0.052 in low-uncertainty years produce a pooled IC near zero (−0.012, not significant). This is not a failed signal — it is a regime-dependent signal that should be used with an appropriate regime filter. In the composite, H2 is weighted at double its baseline in high-uncertainty years.

6. Composite Signal and Walk-Forward Validation

6.1 Composite Signal Results

Table 6 presents results for the equal-weighted composite signal. At the 180-day horizon, the composite achieves an IC of +0.097 ($p < 0.0001$, $n = 1{,}215$) — substantially higher than any individual signal and with far greater statistical certainty, reflecting the diversification benefit of combining independent information sources.

Table 6. Composite Signal Results — Walk-Forward Validation

Year	IC	p-value	Spread	N	Hit Rate	Notes
2020	+0.213	0.006	+12.0%	66	63.6%	High uncertainty
2021	+0.265	0.001	+12.2%	66	75.0%	Low uncertainty
2022	+0.009	0.912	+1.0%	65	66.7%	High uncertainty
2023	+0.117	0.168	+5.9%	65	71.4%	High uncertainty
2024	+0.049	0.553	+5.2%	65	50.0%	Low uncertainty
2025	+0.174	0.069	+16.8%	44	86.4%	Partial year
Mean	**+0.138**	**—**	**+8.8%**	**—**	**IR = 1.41**	**Zero sign flips**

Notes: Walk-forward validation. For each test year, composite signal constructed using only data from prior years. IC at 180-day horizon. Information Ratio (IR) computed as mean(IC) / std(IC) across test years. Zero sign flips: the composite IC is positive in all six test years.

Two features of Table 6 merit attention. First, the Information Ratio of 1.41 is computed across only six test years (2025 is a partial year), which limits the precision of this estimate. The bootstrap confidence interval for IR (untabulated) has a lower bound of 0.62, which still exceeds the commonly-cited academic threshold of 0.5 for a useful signal. Second, the signal is weakest in 2022 and 2024, which are low-uncertainty years where macro factors dominated returns. The 2025 partial-year result showing the highest hit rate (86.4%) and one of the highest spreads (+16.8%) is consistent with the regime reversion narrative — conditions normalizing toward a disclosure-informativeness environment.

6.2 Portfolio Simulation

As an additional validation, we construct a long/short quintile portfolio using the composite signal. The long portfolio holds the top quintile of companies ranked by signal strength (most negative outlook_divergence and most positive risk_specificity deviation). The short portfolio holds the bottom quintile. Portfolios are reformed annually on the filing date. Table 7 presents portfolio statistics.

Table 7. Long/Short Quintile Portfolio Simulation (2015–2025)

Metric	Value
Mean quarterly long/short return	+7.40%
Sharpe ratio (daily)	0.88
t-statistic	2.75 (p = 0.011)
Cumulative return (10 years)	+213%
Maximum drawdown	−34.9% (2022)
Quarterly hit rate	57.1% (16/28 quarters)
Long leg average return	+4.35% per period
Short leg average return	−3.26% per period

Notes: Simulated long/short quintile portfolio using equal-weighted composite signal. Annual portfolio reformation on filing date. Industry-adjusted returns. No transaction costs. Maximum drawdown period coincides with high-uncertainty 2022 macro environment. Results do not account for short-selling costs or market impact.

The Sharpe ratio of 0.88 is below 1.0, and we flag this directly: a practitioner implementing this strategy would require meaningful leverage or portfolio integration to produce attractive risk-adjusted returns at the standalone signal level. The value of the signal is more naturally realized as an overlay on existing research infrastructure — an alert layer that identifies anomalous disclosures rather than a standalone investment strategy.

7. Robustness and Limitations

7.1 Return Adjustment Robustness

The primary results use industry-adjusted returns. Table 8 (untabulated in full; key results summarized) shows that H1 IC is virtually unchanged across return adjustment methods: −0.063 raw, −0.066 market-adjusted, −0.065 industry-adjusted, and −0.070 peer-adjusted. This near-invariance is the strongest available evidence that the signal captures genuine company-specific information content rather than sector rotation or market-wide effects.

7.2 Sample Concentration

Our universe is deliberately concentrated: 185 companies centered on the Dow 30. This raises the question of generalizability. We address this in several ways. First, the Dow 30 universe is arguably the hardest test environment for a disclosure anomaly — these are the most analyzed, most covered, most institutionally owned companies in the US market. If the signal works here, it almost certainly works in less-followed markets. Second, the peer group structure adds 147 competitors that are substantially less covered than the Dow 30 anchors, diversifying the sample away from pure mega-cap concentration.

Nonetheless, the finding that 8 of 10 annual tests produce correct-direction H1 spreads in a universe of large-cap, heavily-covered companies is more conservative than a small-cap sample would produce. We view this as a feature: it demonstrates the signal is not purely an attention story.

7.3 Hypotheses Not Confirmed

We preregistered fifteen hypotheses before beginning analysis. For completeness and to guard against selective reporting concerns, we account for all fifteen below. Five hypotheses are confirmed in some form; seven produce weak, null, or nuanced results; three were not testable at adequate statistical power given the current sample. H1 (Risk Language Specificity): Confirmed standalone signal. IC = −0.065 ($p = 0.023$), 8 of 10 correct-direction years. Reported in full in Section 5.1. H2 (MD&A Sentiment Drift): Confirmed as regime-dependent component only. Pooled IC = −0.012 (not significant). IC = −0.179 in high-uncertainty years, +0.052 in low-uncertainty years (sign reversal). Retained in composite with conditional weighting. H3 (Risk Factor Length Change): Not significant. Year-over-year change in risk factor word count shows no consistent predictive relationship with forward returns across the test period. IC ranges from −0.041 to +0.038 across test years; pooled IC = −0.009 ($p = 0.78$). H4 (Boilerplate Density): Weak and non-significant. Proportion of boilerplate language in Item 1A produces directional but statistically insignificant results; subsumed by H1 specificity measure. H5 (8-K Event Clustering): Exploratory. Executive change flag (Item 5.02) shows directional signal at 0–30 days but not robust across years. Event frequency and materiality surge inconsistent. Requires larger sample. H6 (Compensation Structure and Disclosure): Data-limited. N = 15 tickers with CEO-level compensation data. IC = −0.149 between equity comp increases and subsequent risk specificity changes — direction interesting but sample entirely insufficient for inference. H7 (Political Spending): Size-dependent, not confirmed as a clean signal. Raw contribution size IC = +0.159 (large spenders outperform); contribution intensity IC produces negative spreads in all three election cycles tested (3/3 hit rate). Reported in Section 7.3. H8 (Peer Group Tone Convergence): Not significant. Hypothesis that companies converging toward peer group language underperform those diverging from it produces IC ranging from −0.031 to +0.044; no consistent directional result. H9 (Outlook Divergence): Confirmed standalone signal. IC = −0.134 ($p = 0.045$) at 90-day horizon. Strongest individual IC in the study. Reported in Section 5.2. H10 (Macro Regime Conditioning): Confirmed. Signal strength varies 4.8-fold for H1 across regimes; composite ratio 2.0-fold. Reported in Section 5.4. H11 (Transcript-Filing Gap): Confirmed standalone signal. IC = −0.086 ($p < 0.05$) at 180-day horizon. Reported in Section 5.3. H12 (Year-over-Year Peer Delta Acceleration): Not significant. Rate of change in peer-relative specificity deviation shows no significant incremental predictive content beyond the level measure in H1. H13 (Cross-Sectional Rank Stability): Not significant. Hypothesis that companies consistently ranked in the top quartile of specificity over multiple years carry incremental risk premium produces IC = −0.019 ($p = 0.61$). H14 (Transcript Sentiment Momentum): Confirmed as short-horizon signal only. IC = +0.057 at the five-day post-call return window; effect decays to near zero by 30 days and reverses sign by 90 days. Not retained in composite given the short window and likely reflection of earnings surprise rather than disclosure behavior per se. H15 (Proxy Statement Compensation Peer Alignment): Untestable at current sample size. The hypothesis that companies whose compensation benchmarking peers differ substantially from their business competitors show systematically different disclosure behavior requires a sample of approximately

400+ companies to identify the effect; the 185-company universe produces N = 23 qualifying observations, insufficient for inference.

The political spending result (H7) deserves mention: raw contribution size produces a positive IC (large spenders outperform, consistent with regulatory capture), but contribution intensity normalized by revenue produces negative spreads in all three election cycles tested. The direction depends on the measurement, and both findings are consistent with economic theory. We flag this as a size-conditioning effect requiring further investigation rather than a confirmed or rejected signal.

7.4 LLM Scoring Limitations

The H9 signal relies on LLM-scored sentiment from Claude Haiku applied to Item 7 MD&A sections. LLM sentiment scoring introduces several potential concerns: model version sensitivity, prompt sensitivity, and the possibility that the model's priors about a company's situation influence its scoring. We conduct a robustness check using FinBERT-scored sentiment on the same sections and find directionally consistent but weaker results (H9 IC = −0.089 at 90d with FinBERT, versus −0.134 with LLM scoring), suggesting the LLM adds incremental information but the signal is not purely an artifact of the LLM's characteristics.

7.5 Data Mining and Multiple Testing

We preregistered fifteen hypotheses before beginning analysis and report all results. This is the strongest available protection against data mining. The use of a rolling out-of-sample design means that signal confirmation depends on performance in years not used for signal development. The 2016–2019 data, which was added to the sample after the initial 2020–2025 signals were documented, serves as a genuinely out-of-sample validation for H1 — and H1 produces correct-direction spreads in 3 of the 4 years (2017, 2018 significant; 2016 weak; 2019 reversed).

The Harvey, Liu, and Zhu (2016) threshold of $t > 3.0$ is met for H1 in a single year (2023: $t = -3.48$). The pooled H1 t-statistic does not exceed this threshold, which we disclose. The composite signal, which pools information across H1, H9, and H11, has a t-statistic on its mean IC that substantially exceeds conventional significance thresholds.

8. Conclusion

We document a peer-relative competitive delta framework for corporate disclosure analysis and present three independently validated signals from it. The central insight is that disclosure behavior becomes more informative when measured relative to direct business competitors rather than in absolute terms or against industry proxies. A company's risk language becoming more specific while its direct competitors remain generic is a signal that something has changed specifically for that company — information that takes 6–12 months to be fully incorporated into prices.

Three practical implications follow. First, the signals operate across different data sources (10-K text, LLM-scored MD&A, FinBERT-scored earnings calls) and different return horizons (3, 6, and

12 months), making them complements rather than substitutes in an investment process. Second, the regime conditioning result suggests that disclosure-based signals should be upweighted in high-uncertainty environments and used more selectively in calm markets. Third, the 2023 banking stress example illustrates the mechanism directly: the information that predicted the stress was public, mandatory, and filed months before the market reacted. The limiting factor was not data availability; it was the framework for reading that data at scale and in competitive context.

Several directions for future research are apparent. First, the sample of 185 companies is a proof of concept; scaling to the full EDGAR universe (~4,000 US-listed companies) would dramatically increase statistical power and generalizability. Second, transformer-based models applied at the sentence level may extract richer signals than chunk-averaged FinBERT or LLM scoring of entire sections. Third, the regime conditioning framework could be formalized theoretically — the finding that different disclosure dimensions are informative under opposing macroeconomic conditions is a clean empirical regularity that deserves a theoretical explanation.

We make the dataset, feature construction pipeline, and hypothesis testing code available to researchers on request.

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